LEADHAUL

Human-led autonomous haul convoys for private industrial routes

Pre-seed investment memo · November 2026
Working name · stills and video are simulation renders, not field footage
Figures marked `EST` are our estimates; `TARGET` are plans

The call. Don't build highway convoys. Fuel-only platooning (Peloton) and human-led highway convoys (Locomation, $105M raised) died, and driverless highway trucking is a billion-dollar race. **Pivot to private haul roads** at mines, quarries and heavy-civil sites. Autonomy there is proven and paid for (5,684 autonomous or autonomy-ready trucks, +48% in a year) yet covers only about 3.5% of the fleet. **Wedge:** one operator drives the lead truck and up to three retrofitted trucks follow driverless, with no HD-map survey, control room or new fleet. **Capacity first:** where a site can't staff its trucks, a convoy restores about 240 kt/yr of haulage (≈$1.2M/yr margin at $5/t) for a ≈$0.3M fee. Where it can, a convoy cuts cost per ton by 4–17%, depending on wages. **Ask: $2.5M pre-seed** (post-money SAFE, $18M cap) to a paid driverless-follower pilot in 18 months.

1 Why not the original idea (highway convoys)

- **Fuel savings were too small.** NREL measured 3.7–6.4% team fuel savings, and only ~55–65% of miles can be platooned. Daimler saw "no business case" (2019). Peloton (~$78M raised) was liquidated in 2021.
- **A human leader doesn't shortcut public roads.** A follower in 65 mph traffic needs full driverless validation yet saves half the labor. Locomation had a 1,120-unit commitment and sold its assets in 2023.
- **It is a capital race.** Aurora: ~$3M revenue, $234M loss in Q3 2026. Kodiak burns ~$153M a year; Waabi raised $750M.

2 Where the money is: private haul roads

- **Proven and paid for.** Kodiak runs 37 unsupervised trucks on Atlas's Permian lease roads, with a 100-truck order for 1H27. Volvo hauls all of Brønnøy Kalk's limestone driverless, billed **per tonne** (3 Mt by Sep 2026). Cat had 827 autonomous trucks at end-2025 and is now at Luck Stone quarries; Komatsu passed 1,000.
- **Still early.** The 5,684 autonomous or autonomy-ready trucks (GlobalData, Jul 2026) are ~3.5% of ~160k off-highway rigid trucks and ADTs `EST`. They sit mostly in large single-OEM mines; mid-size, mixed-fleet sites are nearly untouched.
- **Labor is the bottleneck.** 77% of contractors with openings struggle to hire heavy-equipment operators (AGC 2025). About 221k US mining workers, over half the workforce, retire by 2029 (SME). Luck Stone: if one of four trucks has no operator, a quarter of the day's production is gone.
- **Safety and volume.** Powered haulage caused 13 of 33 US mining deaths in 2025 (MSHA). US aggregates were 2.40 Gt ($40.0B) in 2025 from ~3,500 quarries and 6,500 pits (USGS).

3 Product: convoy first, driverless next

Kit. Each follower gets radar and lidar, dual RTK GNSS, a Jetson-class computer, a V2V mesh radio, and by-wire control over J1939 and electro-hydraulic steering. Every truck carries a light lead kit, so any truck can lead and a breakdown drops one truck, not four. Followers drive the leader's V2V breadcrumbs, with offsets that spread wheel-path wear. Gaps are set from stopping distance, each follower checks its own swept corridor, and **any truck can halt the convoy**.

Why convoy first. Followers are driverless either way. What makes it tractable, unlike highways, is the domain: a closed road at ≤35 km/h with trained site traffic. The leader narrows it further: a person drove the route 10–40 s earlier and handles exceptions, so there is no HD map or remote-operations center, and every convoy has a human in it. **Roadmap:** (1) tight convoys on long hauls; (2) path-follow, where followers re-drive the leader's recent path at the loader's pace, with no batching loss; (3) driverless leaders on learned routes.

4 Where convoys win, and where they don't

A convoy removes operator-hours, but its trucks batch at the loader. At 1 min per truck, two loading points and 2% nuisance stops, efficiency is **0.93**. Basis: one convoy, two shifts, $24 per follower-hour, cost per ton including truck ownership.

Site case (per convoy)	Effect	Per year	Break-even eff.
ICP, labor-short (can staff 3 of 4 trucks)	+244 kt hauled	≈$1.2M margin	vs $0.30M fee
ICP, FIFO or union labor ($80/h loaded)	−17.4% cost/t	$526k	0.77
ICP, road-class trucks	−13.8%	$253k	0.80
ICP, path-follow mode (roadmap 2)	−12.7%	$346k	0.86
ICP, remote-site wage $35/h	−8.3%	$216k	0.86
ICP at the US median wage (BLS $28.19/h)	−4.3%	$106k	0.89
Avoid: 12-min quarry loop, one loader	+63% cost/t	efficiency 0.56	0.91

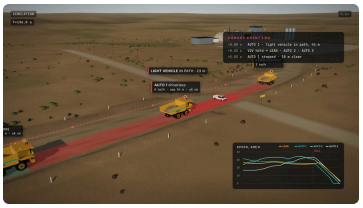
Our ideal customer (ICP): a private haul of at least 3–5 km one-way (travel ≥70% of the cycle), at least 4 trucks, two loading points or a bin loadout, scarce or costly operators, and on-site crossings only. Examples: satellite-pit-to-plant hauls and long mine or megaproject roads. At launch we stay out of short loader-paced loops (OEM and Pronto territory) and frac-sand lease roads (Kodiak's). Followers are billed per ton hauled (≈$0.32/t), so we are paid for production, not hours.



1 One operator leads three driverless followers up an 8.8% ramp.



2 Followers drive the leader's breadcrumbs and check their own corridor.



3 A pickup runs the crossing: the convoy holds over V2V, 18 m clear.



4 Each follower spots and dumps on its own, then forms up again.

5 Business model and unit economics

Robotics-as-a-service. We own the kit, install it and bill per ton hauled by followers. That equals **$24 per follower-hour** at scale TARGET, about 45% of the loaded seat it replaces, with monthly minimums and 12–36-month terms.

Per follower (3,000 billable h/yr)	Revenue	Gross margin	Capex	Payback
Pilot (first 10 kits, $20/h)	$60k	35%	$87k	24 mo
Scale (100+ kits, $24/h)	$72k	67%	$48k	10 mo

Plan TARGET	2027	2028	2029	2030	2031
Sites / followers	pilot	2 / 6	6 / 24	15 / 70	35 / 180
Exit ARR	$0	$0.4M	$1.6M	$5.0M	$13.0M

Market EST : ~160k off-highway rigid trucks and ADTs worldwide (≈16k sold a year × 10-year life). Three followers per four trucks at $72k a year gives a **~$8.6B/yr** TAM; ICP routes are a ~$3.5B SAM. The 2031 plan is 10–20% of our US-aggregates ICP (900–1,800 followers), so it needs mining and heavy civil from 2029, and about $40M of equity.

Go-to-market. Land one convoy on one route, then expand across the site's routes and the owner's sites. **Buyers:** site GMs and operations VPs at multi-site aggregates and mining companies, and contract haulers paid per ton. **Channels:** ADT dealers and lessors who carry kit capex (as Holman does for ISEE). Expect 6–12 months to a paid pilot EST.

6 Competition and why we win the wedge

Player	Model	Our angle
Cat, Komatsu AHS	OEM fleets with site infrastructure; big mines, now quarries	Mixed fleets, no site-wide infrastructure, mid-size sites
Pronto (Atoms) + SafeAI	Retrofit driverless; ~30 vehicles at Heidelberg	Human-led: narrower safety case, long hauls in mixed traffic
ASI ($225M SoftBank), Bedrock	Generalist construction autonomy	Convoy coordination; partner, e.g. haul behind their loaders
Kodiak	Driverless new trucks on frac-sand lease roads	Not its lane at launch; retrofit fleets that won't buy new trucks
Teleo/Havoc, Fernride	Remote supervision	No remote operations center; works where links are poor

Could an OEM or Pronto add a follow mode? Yes. Our defense is speed, mixed-fleet retrofits and the coordination stack (convoying, spacing, merging, spotting), plus route data and an ISO 23725-aligned API into incumbent fleet systems. They are also the natural acquirers: SafeAI went to Pronto, Teleo to Havoc, ASI Mining to Epiroc.

7 Technical plan: 18 months, gated

Already built. The founder's multi-vehicle MPC/RL team-driving stack (follow mode, gap rules, CARLA bridge) has 0 crashes on 50 seeds in 45-car traffic. Our haul-convoy sim (rigid 60 t trucks) tracks the operator's line to **1.7 cm RMS** over 20 Hz V2V with 40–60 ms latency and 2% loss (field target ≤15 cm). It held for a pickup that ran a crossing 44 m ahead and stopped 18 m clear.

Months	Milestone (gate)
0–3	ICP list from MSHA mine data and satellite haul lengths; 40 interviews; 3 LOIs; HARA, ODD, safety-case skeleton (ISO 17757, UL 4600)
3–6	J1939 hardware-in-the-loop bench; 3 by-wire UTVs convoying with fault injection. **Gate 1:** sim-vs-real residuals; partner and truck chosen
6–12	Retrofit 1 lead + 2 followers at the partner's isolated loop; 300+ supervised convoy-hours. **Gate 2:** efficiency ≥0.93; nuisance-stop and intervention rates. Seed raise starts
12–18	Driverless followers in an isolation zone (Freeport Bagdad playbook); 500+ driverless hours; third-party pre-audit; paid pilot (≥$15k/month) to a 12-month contract; second-site LOI

Safety and approvals. Simplex design: MPC and frozen learned policies drive; a simple verified controller and an independent e-stop own the minimal-risk stop. Each gap is at least the predecessor's stopping distance plus margin (a loaded truck stopping downhill can need 60–120 m). A comm-loss watchdog stops any follower that loses its link, and radar leads in dust. MSHA has no autonomy pre-approval: sites document autonomy in their surface mobile equipment safety program, reviewed at inspection. We carry product liability; contracts allocate operational responsibility.

8 The raise

$2.5M pre-seed · post-money SAFE · $18M cap (~13.9%), at Carta's Q2 2026 median cap for $1–2.4M rounds.
Use (18 mo): team of founder, 3 engineers and a field-ops/mining lead $1.27M · by-wire partner and OEM access $0.19M · 2 ADTs + 3 UTVs $0.30M · 3 kits $0.23M · safety case, audit, insurance $0.25M · legal/IP and discovery $0.10M · contingency $0.17M.
Seed: raise from month 12, ~$8M at $28–35M post, on a paid pilot moving to a 12-month contract, 500+ driverless hours with falling interventions, a safety pre-audit and a second-site LOI.

Team. The founder brings convoy control and multi-vehicle MPC/RL. Before close: a field-ops/mining co-founder and 1–3 site LOIs. **Investors:** Eclipse (led Bedrock's and Humble's seeds), 8VC (Overland, Outrider), Zacua (new $135M mining and construction robotics fund), Schematic, Trucks VC, Root, Construct, DCVC; strategic Samsara Ventures; Point72 or Shield for defense. **Non-dilutive:** state mobility grants (Michigan gave Intramotev $200k) and convoy modules for ATV-S primes. SBIR authority lapsed in Oct 2025; check whether it has been renewed.

9 Risks, mitigations, kill criteria

Risk	Mitigation / kill criterion
Too few ICP sites	Sized in the first 90 days. **Re-pivot** if there are fewer than 25 qualifying sites, or fewer than 3 willing to pay for a pilot, by month 6
Convoy efficiency below break-even (≈0.86)	Measured at Gate 2. If it misses, lead with path-follow mode, or license the convoy stack to OEMs and integrators
OEMs or Pronto add a follow mode	Speed, mixed fleets, ISO 23725 integration, dealer and contract-hauler channels; they are also the likely acquirers
Safety incident, nuisance stops	Isolation zone first, simplex architecture, stopping-distance gaps, lead kit on every truck, third-party audit, insurance
Retrofit and labor friction	One J1939 truck model and a by-wire partner first. The operator becomes the paid convoy lead, with relief and escort roles (USW at Asarco)

Sources, with links for every figure, are in `venture/research/sources.md`. Key ones: NREL (2015, 2018), FreightWaves and Jones Day on Locomation, Aurora and Kodiak Q3 2026 results, Volvo Group, Caterpillar, Komatsu, GlobalData, USGS MIS 2026-Q2, BLS OOH and ECEC, AGC 2025, SME, MSHA, Carta Q2 2026, and company releases. Model: `venture/model/unit_economics.py` (wages, truck costs, efficiency and margins are stated assumptions). Simulation: `venture/sim/convoy_sim.py`. Fleet counts and ICP prevalence are estimates still to be validated.